LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	DEEKSHITHA U
Register Number	192424055
Date	10 October 2026
Day	Saturday

SET 18: Product Inventory Management

Aim

To visualize product inventory using a bar chart, stacked bar chart, table, and an interactive Tableau dashboard.

Dataset: Product Inventory

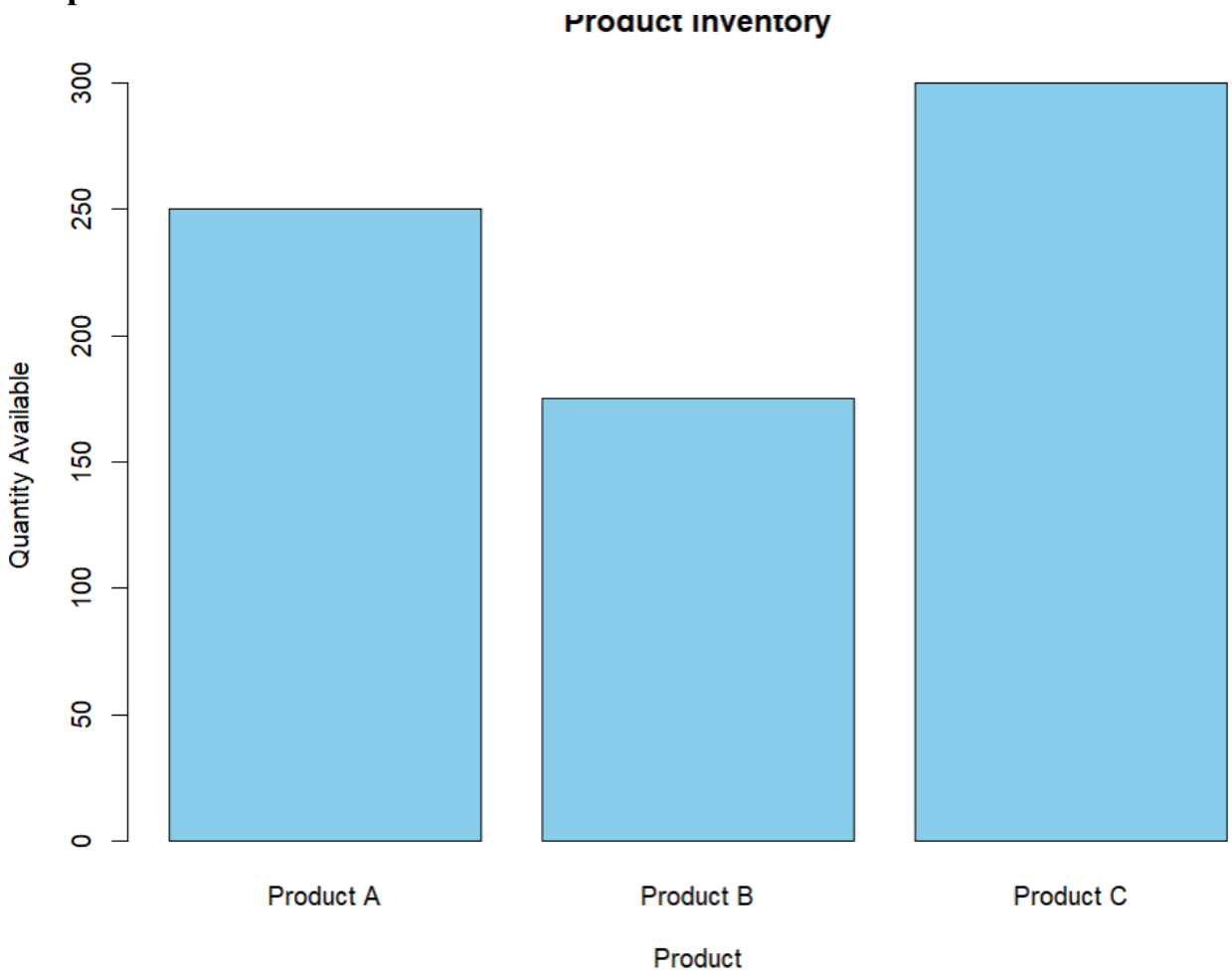
Product ID	Product Name	Quantity Available	Price ($)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Q1. Create a bar chart to visualize the quantity of each product in the inventory.

R Program

```
product <- c("Product A", "Product B", "Product C")
quantity <- c(250, 175, 300)
barplot(
  quantity,
  names.arg = product,
  xlab = "Product",
  ylab = "Quantity Available",
  main = "Product Inventory",
  col = "skyblue"
)
```

Output:



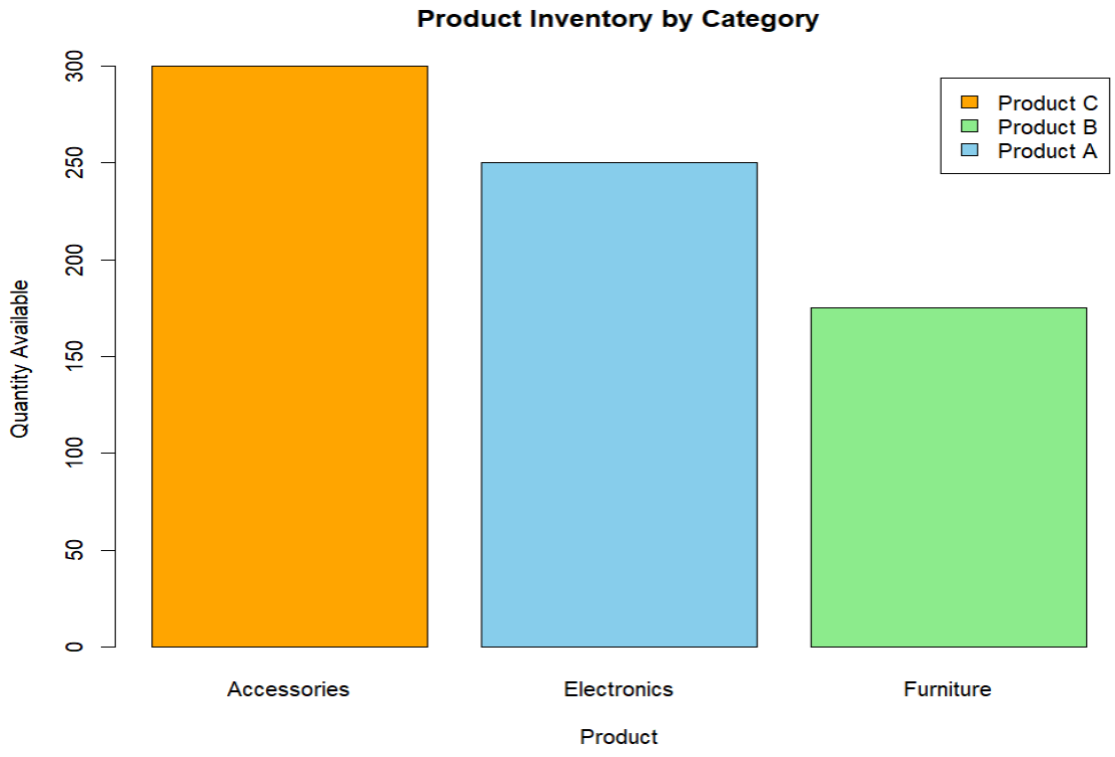
Observation: Product C has the highest inventory with 300 units, while Product B has the lowest with 175 units.

Q2. Generate a stacked bar chart to show the quantity of each product within different product categories.

R Program

```
product <- c("Product A", "Product B", "Product C")
category <- c("Electronics", "Furniture", "Accessories")
quantity <- c(250, 175, 300)
inventory_table <- xtabs(quantity ~ product + category)
barplot(
  inventory_table,
  beside = FALSE,
  xlab = "Product",
  ylab = "Quantity Available",
  main = "Product Inventory by Category",
  col = c("skyblue", "lightgreen", "orange"),
  legend.text = TRUE
)
```

Output:



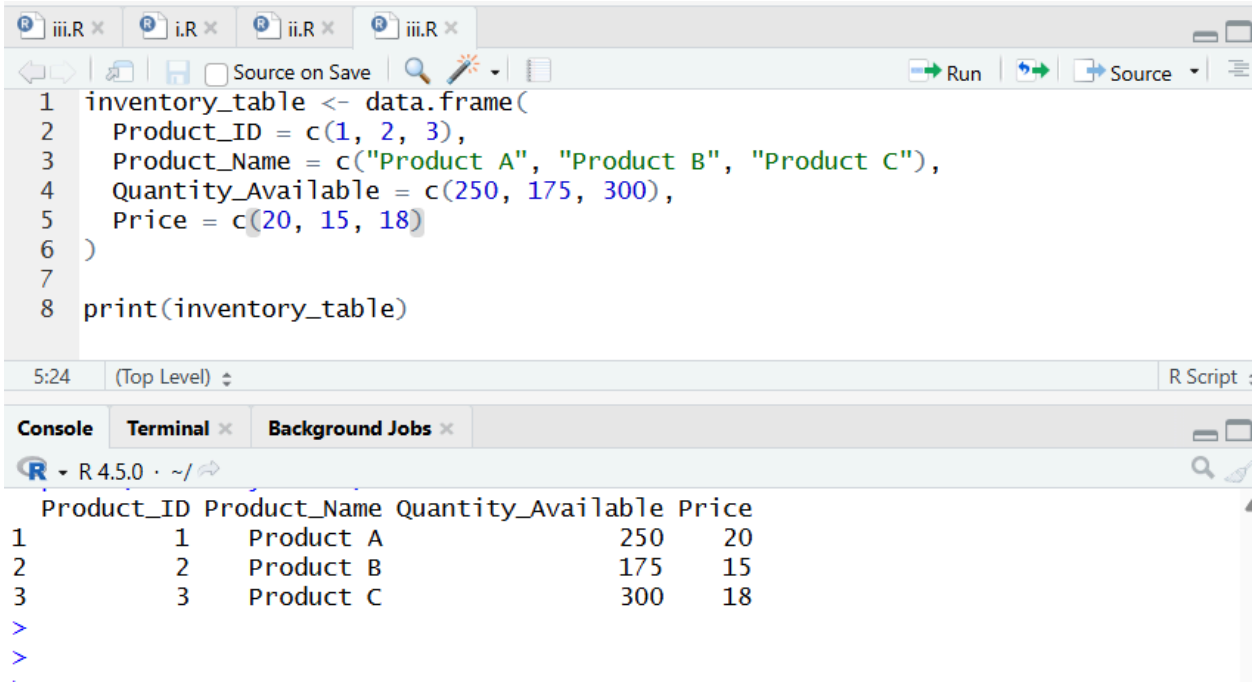
Observation: Each product belongs to a different illustrative category, so each bar contains only one category segment. Actual category information is needed for a meaningful multi-segment stacked chart.

Q3. Build a table to show the inventory data for each product.

R Program

```
inventory_table <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)
print(inventory_table)
```

Output



Observation: The table displays each product's ID, name, available quantity, and price.

Q4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and table.

Step 1 — Prepare the CSV dataset

Save this as Product_Inventory.csv.

Product ID,Product Name,Quantity Available,Price
1,Product A,250,20
2,Product B,175,15
3,Product C,300,18

Step 2 — Import the dataset

1. Open Tableau.
2. Select Connect → Text File.
3. Import Product_Inventory.csv.
4. Open a new worksheet.

Step 3 — Create the bar chart

1. Create a new worksheet named Inventory Bar Chart.
2. Drag Product Name → Columns.
3. Drag Quantity Available → Rows.
4. Select Bar from the Marks dropdown.
5. Drag Quantity Available → Label.
6. Set the title to Product Inventory.

Step 4 — Create the stacked bar chart

First, add a Category column to your CSV using the illustrative categories shown in Q2.

1. Create a new worksheet named Inventory by Category.
2. Drag Product Name → Columns.
3. Drag Quantity Available → Rows.
4. Drag Category → Color on the Marks card.
5. Select Bar from the Marks dropdown.
6. In the top menu, select Analysis → Stack Marks → On.
7. Set the title to Product Inventory by Category.

Step 5 — Create the inventory table

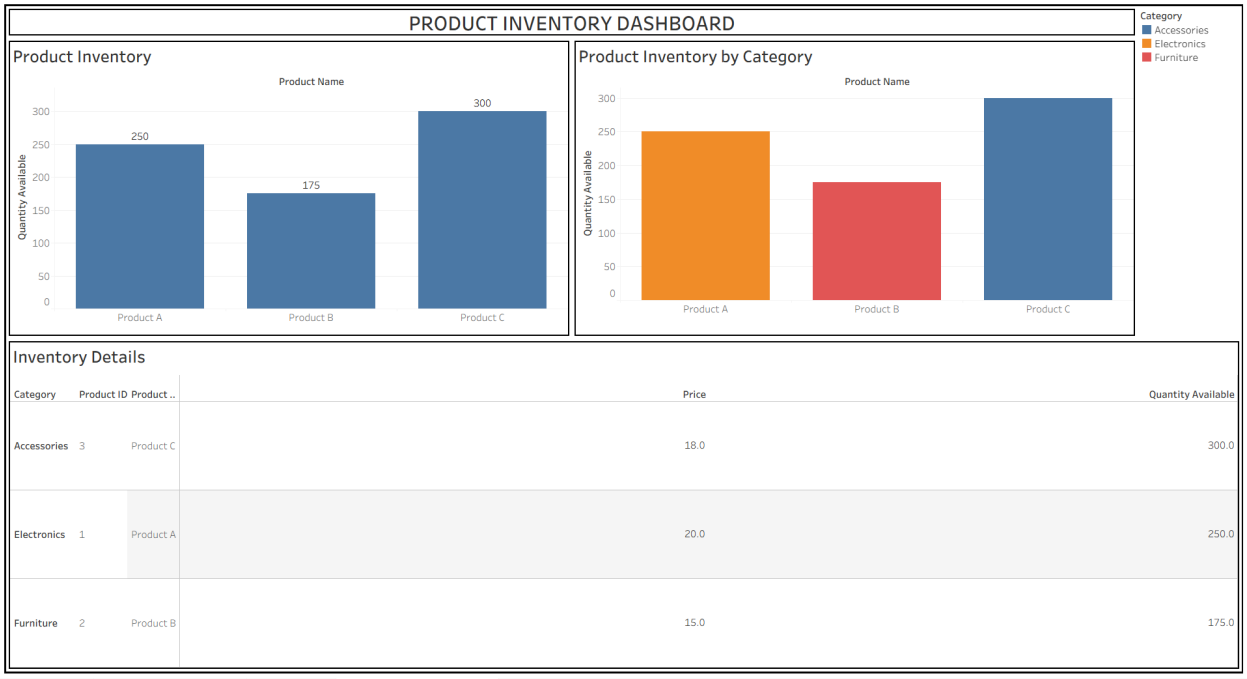
1. Create a worksheet named Inventory Details.
2. Drag Product Name → Rows.
3. Add Product ID, Quantity Available, and Price to the view.
4. Select Show Me → Text Table if necessary.
5. Set the title to Product Inventory Details.

Step 6 — Create the dashboard

1. Click New Dashboard at the bottom.
2. Add the Inventory Bar Chart worksheet.
3. Add the Inventory by Category worksheet.
4. Add the Inventory Details worksheet.

5. Arrange both charts at the top and the table below.
6. Add a Category filter if required.
7. Set the dashboard title to Product Inventory Dashboard.

Expected dashboard



Procedure

1. Entered the product inventory dataset.
2. Created a bar chart, stacked bar chart, and inventory table using R.
3. Imported the inventory dataset into Tableau.
4. Created individual worksheets for the visualizations.
5. Combined the worksheets into a dashboard for interactive exploration.

Result

The product inventory was visualized using bar charts and a table. A Tableau dashboard combining inventory quantities, category information, and product details was designed for interactive exploration.